LAB 05 - INDIVIDUAL EVIDENCE SHEET

Technical Document Preparation with LaTeX

Name	Abdur Rehman	**Roll No.**	623486
Group	13	**Lab 03 Topic**	Expectation vs Reality: AI in Education

Complete this sheet progressively during the lab. Record evidence from your own project and be ready to demonstrate any item live. The final PDF alone is not sufficient evidence of learning.

A. Source, Compilation, and Document Structure

Evidence prompt	Your evidence
Main source file	main.tex
First successful compilation	[ ] Overleaf Time/attempt: 6th October 4:29pm
Difference between source and output	.tex is: the source code/markup in latex PDF the output/build file
One section/subsection I personally created	**AI Benchmarks in the methodology section**
Starter-structure check	[ ] documentclass [ ] preamble [ ] begin/end document

B. Figure, Table, and Cross-References

Evidence prompt	Your evidence
Figure filename/path: graph.png	
Figure label	Fig: Bar Chart
Table label	Tab: model_analysis
One figure/table reference used in text	66 67 Particularly Deepseek which none of the respondents seemed to use as can be seen in 68 \textbf{Table~\ref{tab:model_analysis}} and \textbf{Figure~\ref{fig:Bar Chart}}
Why use \label + \ref instead of typing Figure 1/Table 1?	**In case you add more tables/ labels, their reference numbers will automatically be updated**

C. Equation and Reference

Evidence prompt	Your evidence
Equation label	eq:model_percentage
Where the equation is referenced	75 Using the model Percentages (See Equation \ref{eq:model_percentage} for details.)
If a reference displays ??, what should you check/do first?	**Check if label is defined or misspelled in reference**

D. Bibliography and Citations

Evidence prompt	Your evidence
Bibliography filename	references.bib
Number of legitimate references	4 (target: 4-5)
One citation key	`@techreport{berkeley2025math, author = {Berkeley EECS Technical Reports}, title = {Benchmarking Large Language Models on Advanced Mathem institution = {University of California, Berkeley}, type = {Technical Report}, number = {EECS-2025-121}, year = {2025}, url = {https://www2.eecs.berkeley.edu/Pubs/TechRpts/2025/Arc }`
Where I used \cite{...}	`\subsection{Project Overview}` `\section{Project Overview}` People have many grievances about AI when used for learning: wrong outputs, hallucinations, and lengthy gibberish are among a few \cite{gokcearslan2024chatbots}. We decided to ask people about common problems they face when using AI (particularly LLMs) in educational settings \cite{chen2020ai} and then Benchmark the Most commonly used models against AMC Maths Problems \cite{berkeley2025math}. Doing so allows us to investigate if there exists a gap between actual benchmark results and users' predicted results \cite{katsantonis2024students}.
What connects \cite{key} to the reference list?	**The citation key / labels in the references.bib file**

E. Overleaf to Local LaTeX Workflow

Evidence prompt	Your evidence
Local editor	**Latex workshop vs code extension + basictex macos package**
LaTeX distribution/compiler environment	**lualatex/basictex**
Local compilation	[ ] Successful [ ] Needed troubleshooting
One change I made locally	**Made the title bold**
Key idea	Overleaf is: an online editor LaTeX is: the medium of development / markup language

F. Troubleshooting Record

Evidence prompt	Your record
Error/symptom observed	**Table not formatted correctly (extra column)**
Likely cause	**Table structure may have included an extra 'c' for column**
Correction made	**Remove don 'c' from '|l|c|c|c|'**
Troubleshooting sequence	[ ] Read [ ] Locate [ ] Interpret [ ] Correct [ ] Recompile

G. Word/Office vs LaTeX

Concept	Office / Word	LaTeX
Heading	Heading Style	\section{}
Caption	Insert Caption	\caption{}
Cross-reference	Cross-reference tool	\label{} + \ref{}

Citation	Citation manager	\cite{} + .bib
Main editable file	.docx	.tex

H. Individual Exit Task / Viva

Item	Record
Task assigned	[ ] Structure [ ] Figure [ ] Table [ ] Equation [ ] Citation [ ] Error fix
What I was asked to do	n/a
Result / explanation	n/a
Instructor / lab engineer initials	QA (Quindeel Fatima)

I. External Assistance Disclosure

[ ] None [ ] Official documentation/search [ ] AI clarification/troubleshooting [ ] Peer help

Analytic Evaluation Rubric - 10 Marks

Use the descriptor that best matches the demonstrated performance and award marks within the stated criterion weight. Half marks may be used where appropriate. A polished group artifact does not replace individual live competence; live demonstration/viva marks must reflect the individual student's performance.

Criterion	Weight	Excellent	Good	Satisfactory	Needs Improvement
Document structure & compilation	1.5	Creates a valid LaTeX document with correct preamble/body, title metadata and successful compilation; clearly explains .tex source vs generated PDF.	Document compiles and structure is mostly correct with minor syntax/organization issues.	Basic document compiles with assistance; structure understanding is partial.	Document does not reliably compile or student cannot identify/explain core LaTeX structure.
Sections, formatting & lists	1.0	Uses sections/subsections, paragraphs, emphasis and list environments correctly and consistently to structure Lab 03 content.	Required structures are present with minor syntax/organization issues.	Basic structure is present but inconsistent or created with substantial prompting.	Major required structures are missing or syntactically incorrect.
Table & figure construction	1.5	Creates a correct table and figure using appropriate environments, real Lab 03 data/chart, meaningful captions/labels and correct image path.	Table/figure are functional with minor formatting, caption, label, or path issues.	Both are present but one requires assistance or contains notable structural problems.	Table/figure are missing, fail to compile, or are not understood.
Cross-referencing	1.5	Correctly labels and cross-references figure, table and equation using meaningful labels and explains automatic numbering/recompilation.	Most references work correctly; minor label/reference issues remain.	Some cross-references work but student needs prompting or uses inconsistent/manual numbering.	References are missing/broken or student relies mainly on manually typed numbers.
Equation typesetting	0.5	Includes a meaningful numbered equation and references it correctly in the report.	Equation is correct with a minor formatting/reference issue.	Equation is present but needs assistance or is weakly integrated.	Equation is missing, incorrect, or cannot be compiled/explained.
References, citations & bibliography	1.5	Uses a separate .bib file with 4-5 legitimate references, correct citation keys/in-text citations and a successfully generated bibliography.	Bibliography workflow works with minor metadata/citation issues.	References are present but citation/BibTeX workflow requires substantial assistance or has unresolved issues.	References are invented/missing, .bib workflow fails, or citations cannot be explained.
Overleaf + local workflow	0.5	Successfully transfers project, locates all dependencies, compiles locally, makes a local edit and explains that Overleaf is an environment rather than LaTeX itself.	Local workflow succeeds with only a minor issue or limited explanation.	Local compilation succeeds only with assistance or project transfer is incomplete.	Cannot transfer/compile locally or misunderstands the relationship between Overleaf and LaTeX.
Individual evidence, troubleshooting, exit task & viva	2.0	Evidence is complete; student independently edits source, diagnoses a syntax/path/reference/citation issue, recompiles, and explains relevant code in own words.	Demonstrates individual competence with minor prompting and mostly correct explanations.	Completes basic task with significant prompting; troubleshooting/explanation is limited.	Cannot independently modify/troubleshoot the source or explain submitted LaTeX work.